

Development of CNC Wire Bending Machine and Software for Real Time Portable Art Fabrication

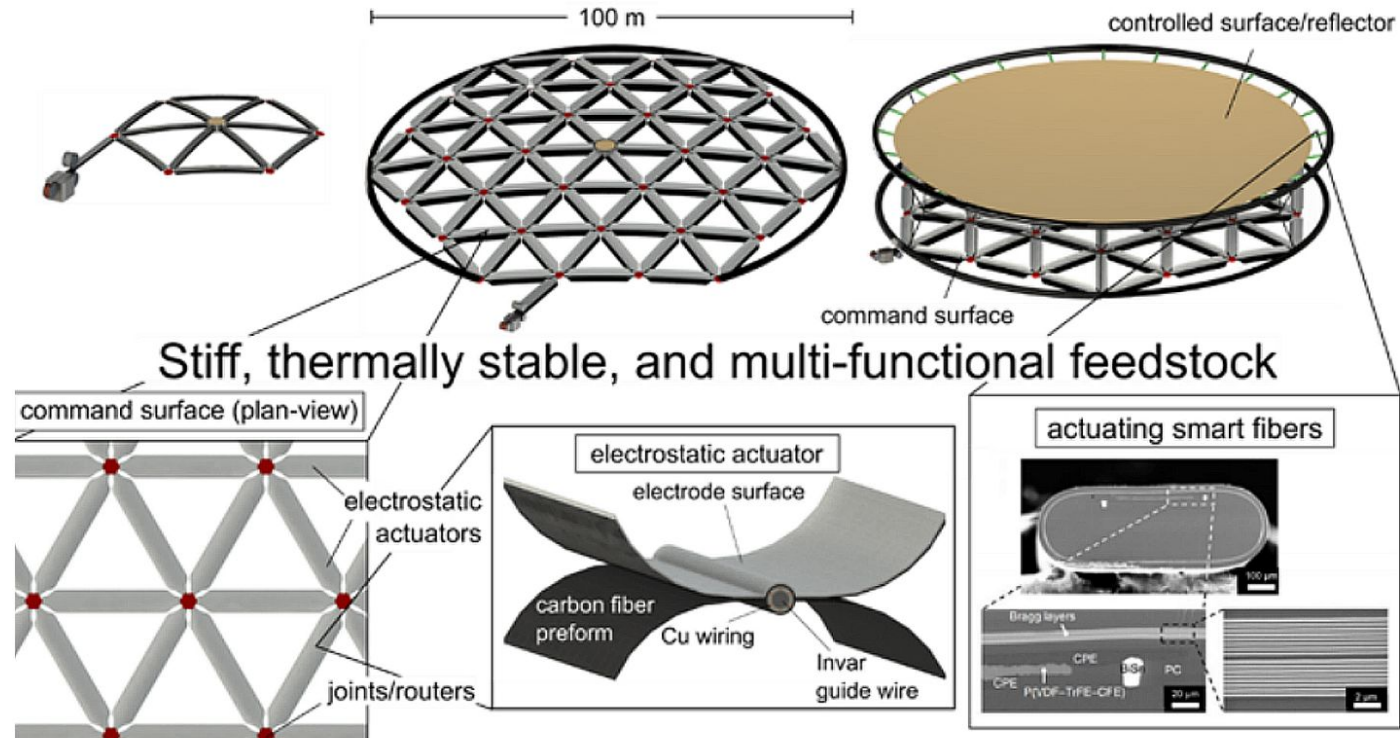
Horatio Montero Hamkins



Supervisor: Dr. Evros Loukaides
Assessor: Dr. Michael Wilson

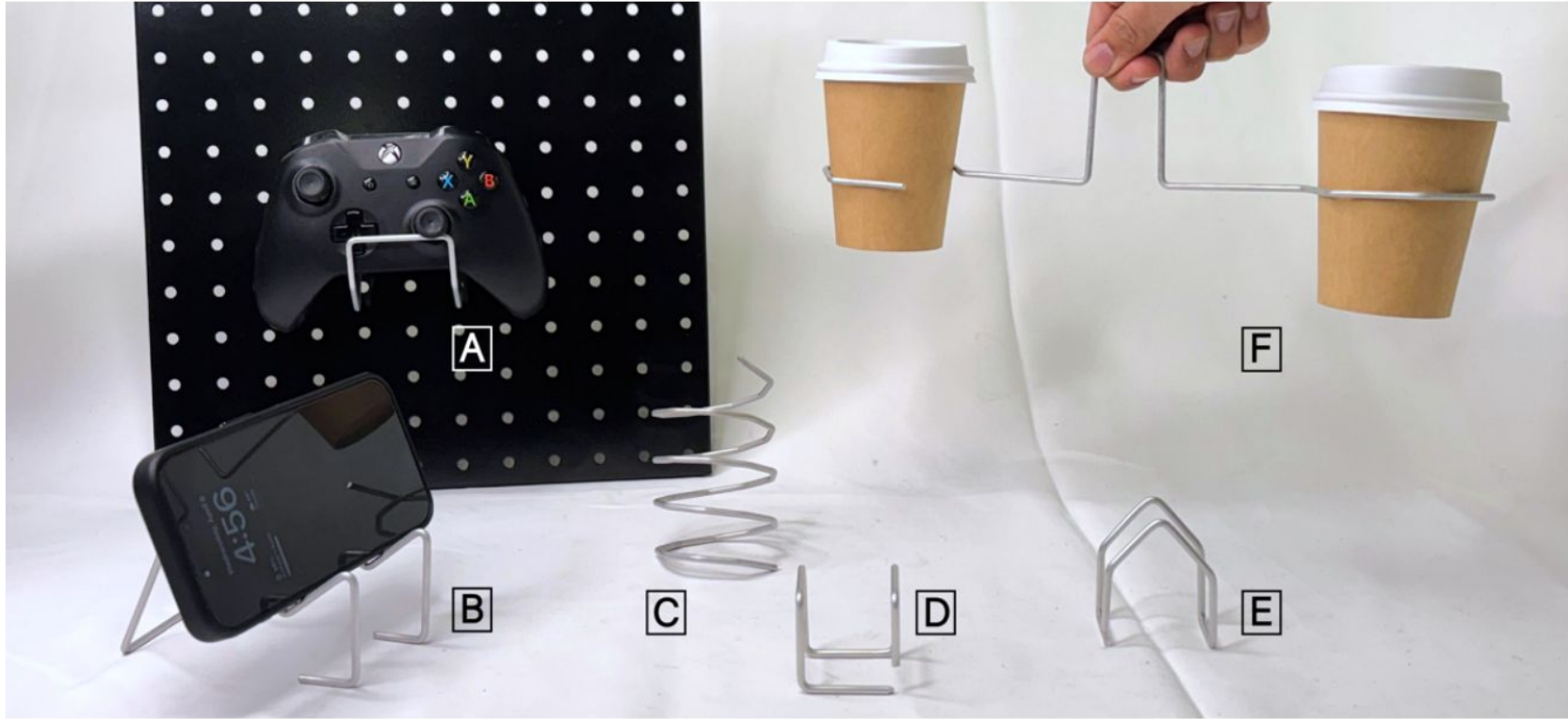


Motivation: In-Orbit Assembly



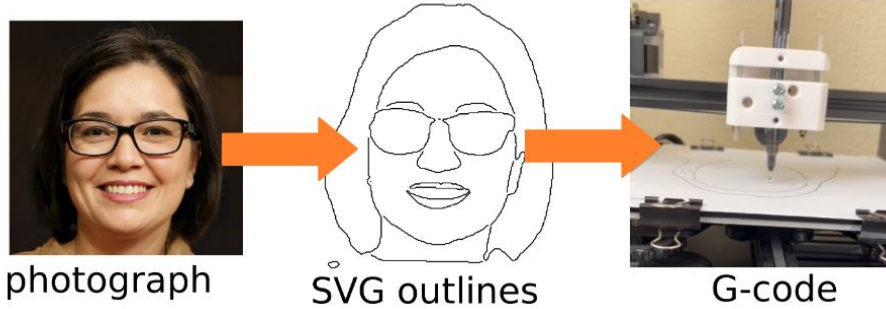
Bend-Forming of Large Electrostatically Actuated Space Structures (Cordero 2024)

Motivation: Enabling Personal Fabrication

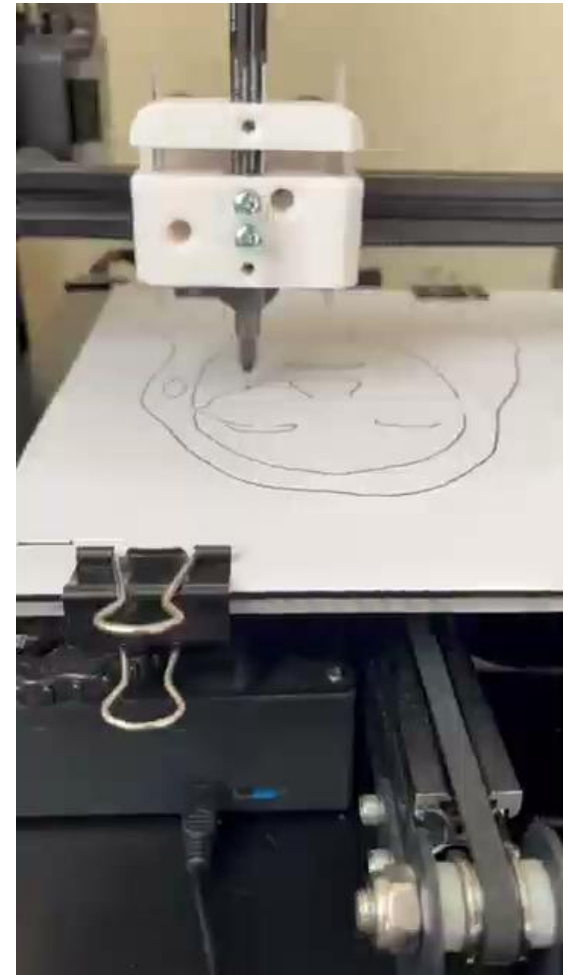


WireBend-kit: A Computational Design and Fabrication Toolkit for Wirebending Custom 3D Wireframe Structures (Faruqi et al., 2025)

Background: Portrait-Outliner



github.com/horatioh13/Portrait-Outliner (own work)



Motivations: Art Fabrication



AI-Da's drawings are fed into AI algorithms to create abstract paintings.

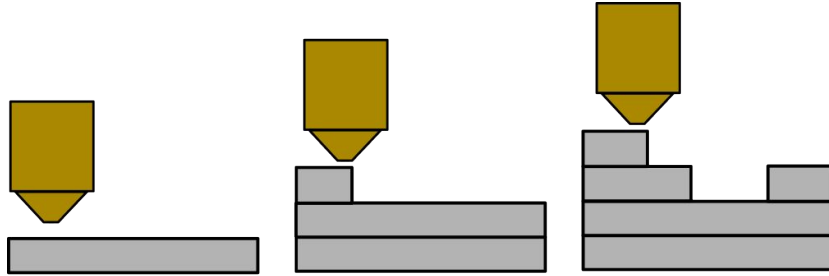
Photo by Nicky Johnston



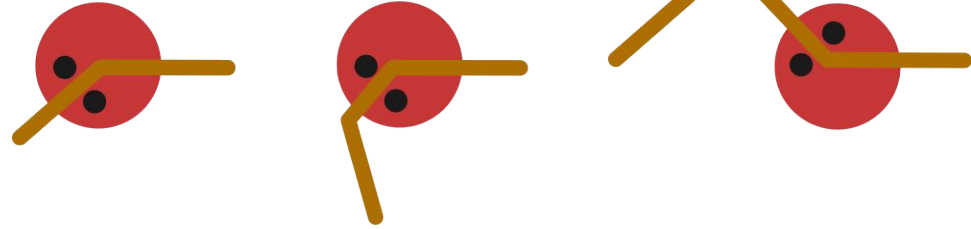
AI-Da testifying in the house of lords on her 'creative' process.

Photo by Roger Harris

Motivations: Rapid Prototyping



Additive Manufacturing
builds objects voxel by voxel

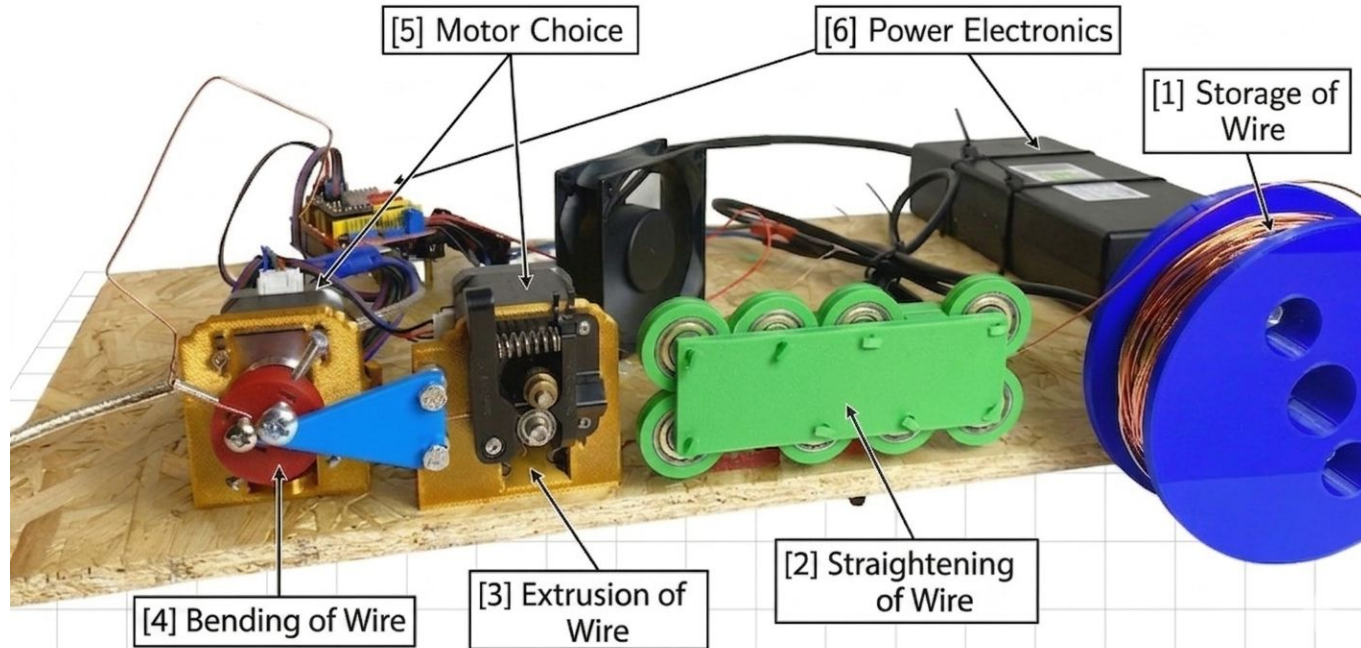


Formative Manufacturing
Changes an Objects Shape

Formative manufacturing is 10x quicker than additive manufacturing,
as 3D printing fabrication time grows with the cube of the size of the object.

Aims and Objectives

Objective 1: The creation of a reliable, high-speed, energy efficient wirebending machine

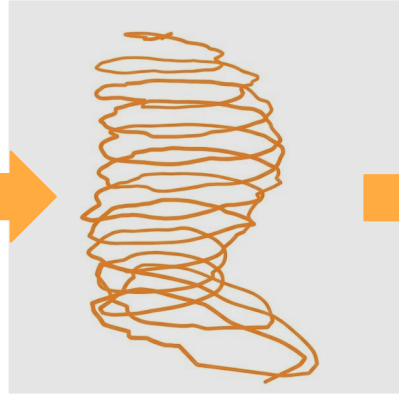


Aims and Objectives

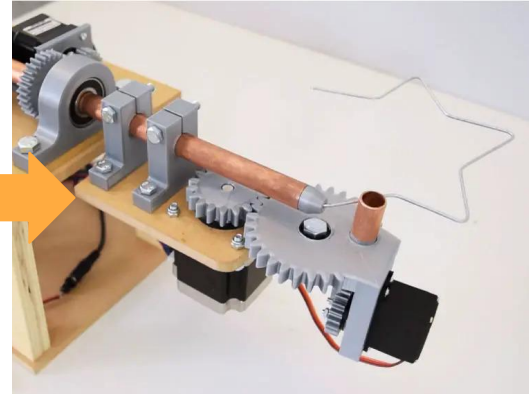
Objective 2: The creation of custom slicing software



3D model



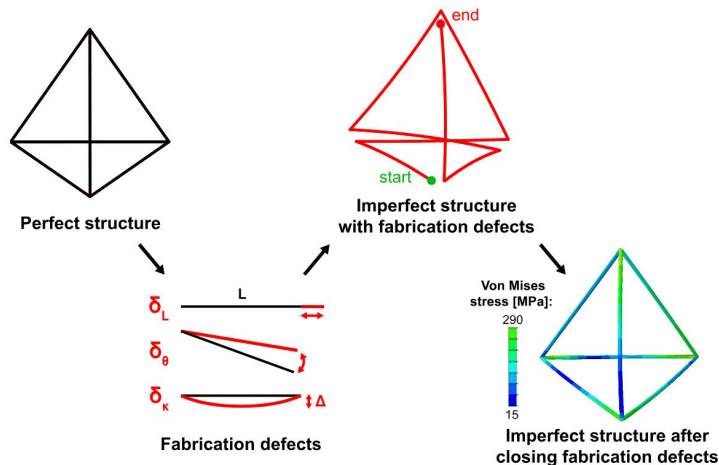
contour generation



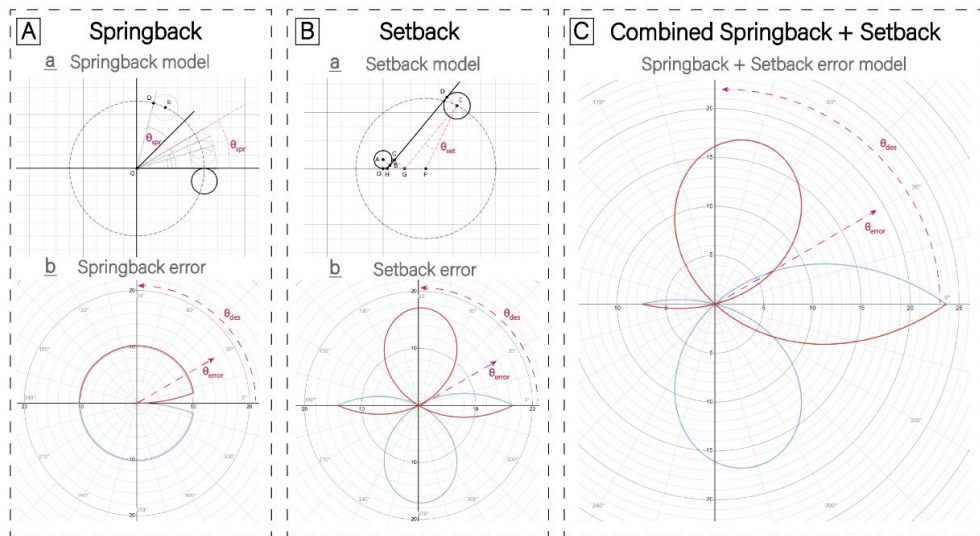
G-code

Aims and Objectives

Objective 3: Analyse the limitations of the wirebending machine and software algorithms

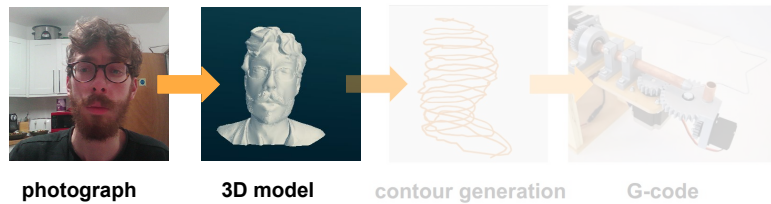


Bend-Forming: A CNC deformation process for fabricating 3D wireframe structures (Cordero 2023)

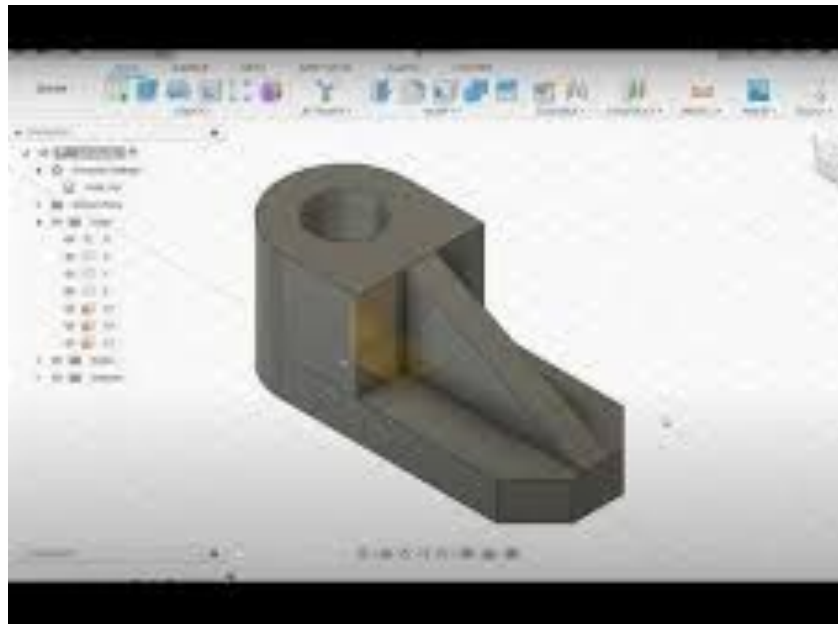
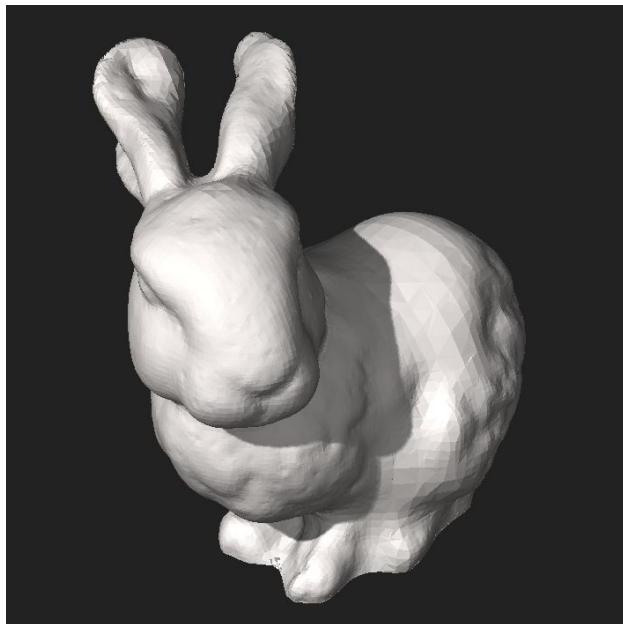


WireBend-kit: A Computational Design and Fabrication Toolkit for Wirebending Custom 3D Wireframe Structures (Faruqi et al., 2025)

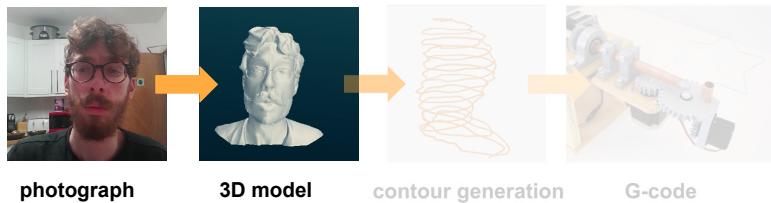
Methods



User-provided CAD model



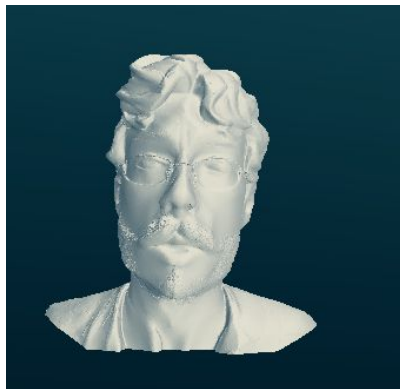
Methods



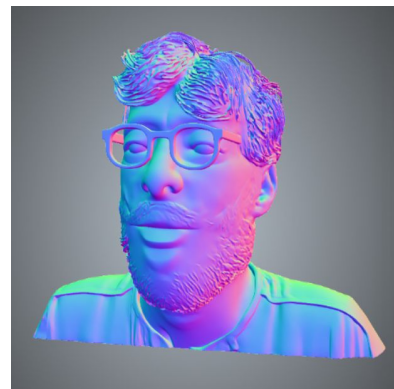
2D - 3D Generative Models



Hunyuan3D-2
(open source)

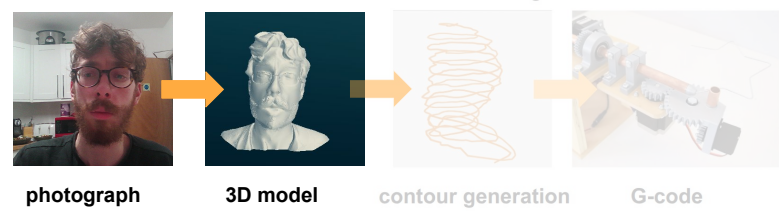


Bambulabs Make My Statue



Rodin AI

Methods



Text - 3D generative model

user query → semantic string matching → return best scoring object

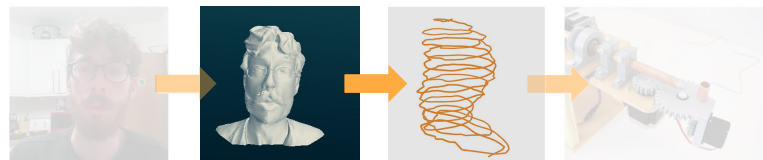
“Teapot”
“Fighter jet”
“Old computer”



object database (1,814 models)



Methods

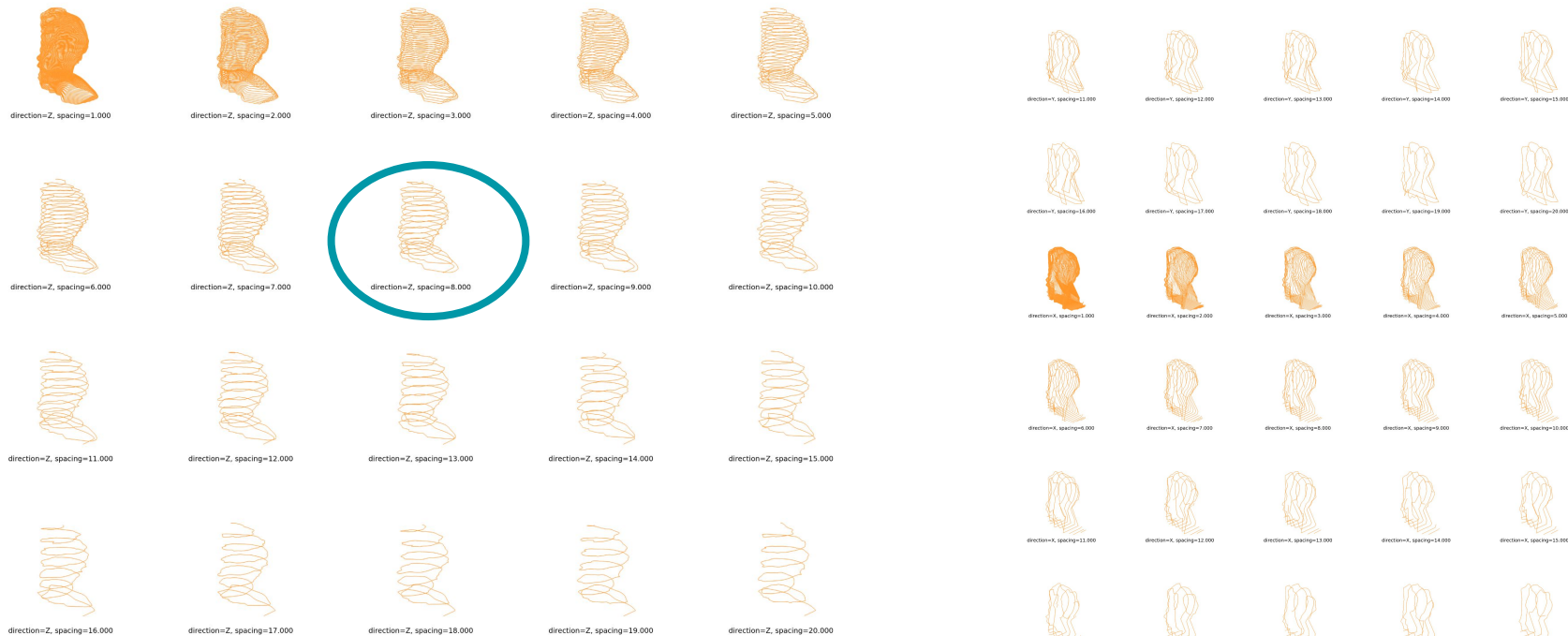


photograph

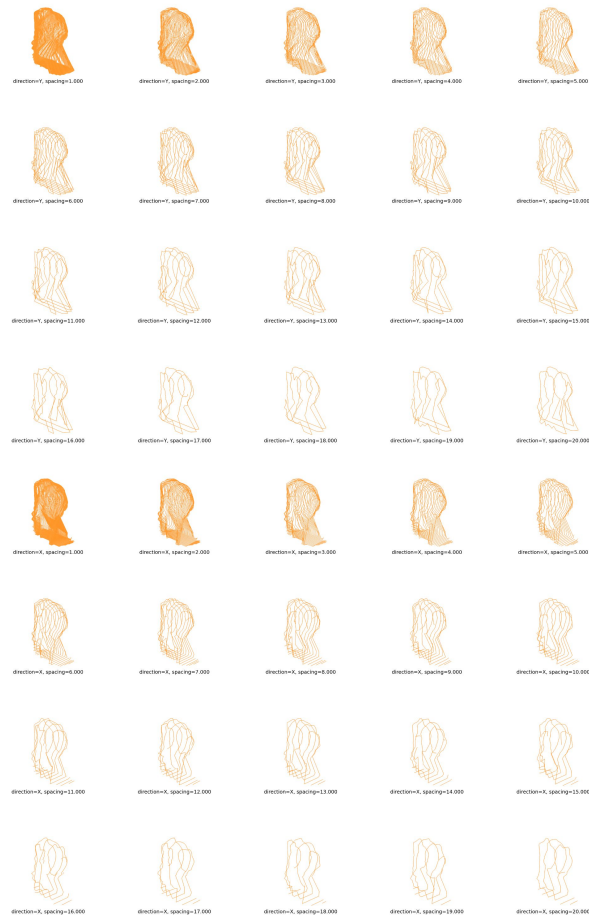
3D model

contour generation

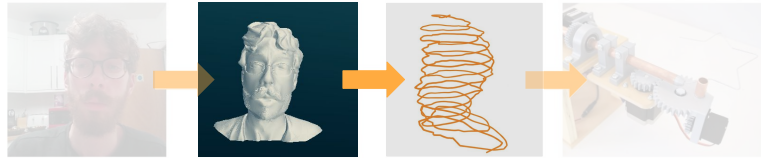
G-code



Option 1: contour generation



Methods

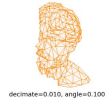


photograph

3D model

contour generation

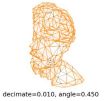
G-code



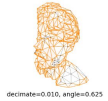
decimate=0.010, angle=0.300



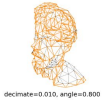
decimate=0.010, angle=0.275



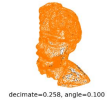
decimate=0.010, angle=0.450



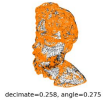
decimate=0.010, angle=0.625



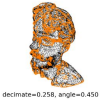
decimate=0.010, angle=0.800



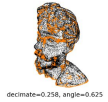
decimate=0.258, angle=0.100



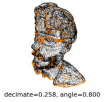
decimate=0.258, angle=0.275



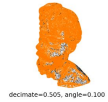
decimate=0.258, angle=0.450



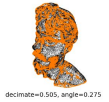
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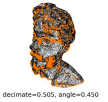
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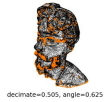
decimate=0.505, angle=0.100



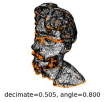
decimate=0.505, angle=0.275



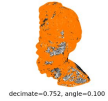
decimate=0.505, angle=0.450



decimate=0.505, angle=0.625



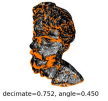
decimate=0.505, angle=0.800



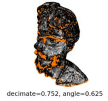
decimate=0.752, angle=0.100



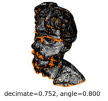
decimate=0.752, angle=0.275



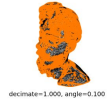
decimate=0.752, angle=0.450



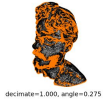
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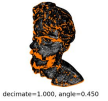
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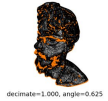
decimate=1.000, angle=0.100



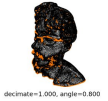
decimate=1.000, angle=0.275



decimate=1.000, angle=0.450



decimate=1.000, angle=0.625



decimate=1.000, angle=0.800

Option 2: Edge Detection



decimate=0.001, angle=0.500



decimate=0.001, angle=0.625



decimate=0.001, angle=0.750



decimate=0.001, angle=0.875



decimate=0.001, angle=1.000



decimate=0.003, angle=0.500



decimate=0.003, angle=0.625



decimate=0.003, angle=0.750



decimate=0.003, angle=0.875



decimate=0.003, angle=1.000



decimate=0.006, angle=0.500



decimate=0.006, angle=0.625



decimate=0.006, angle=0.750



decimate=0.006, angle=0.875



decimate=0.006, angle=1.000



decimate=0.008, angle=0.500



decimate=0.008, angle=0.625



decimate=0.008, angle=0.750



decimate=0.008, angle=0.875



decimate=0.008, angle=1.000



decimate=0.010, angle=0.500



decimate=0.010, angle=0.625



decimate=0.010, angle=0.750

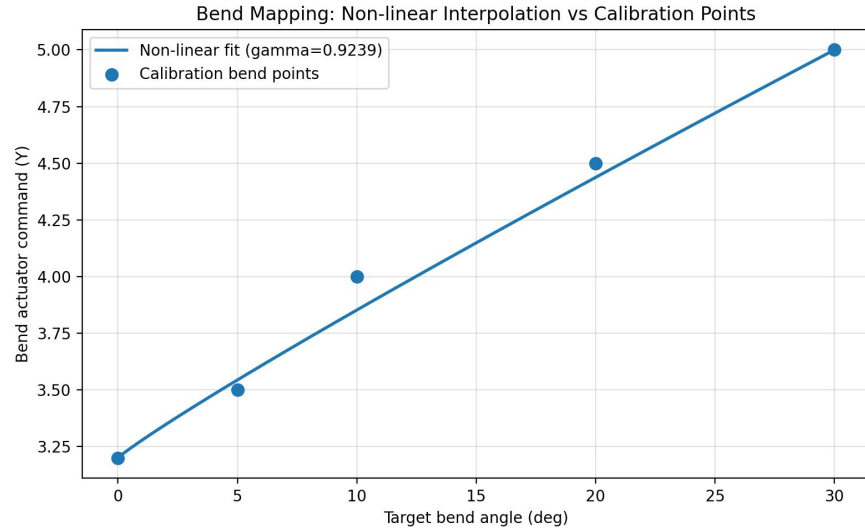
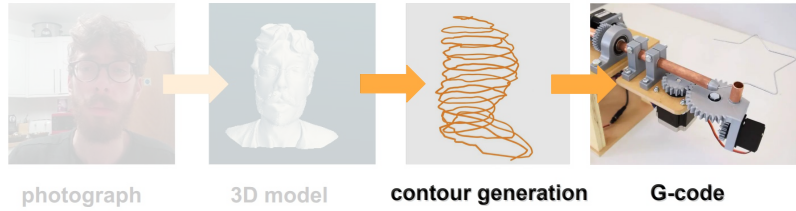


decimate=0.010, angle=0.875



decimate=0.010, angle=1.000

Methods



Conversion of target bend angle to G-Code Instructions

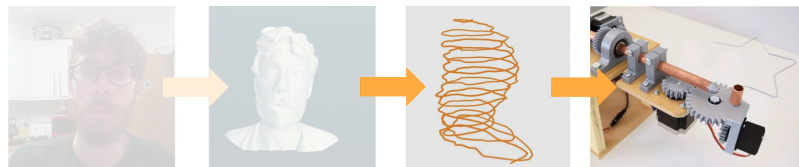
```
F22.0  
B33.1  
F3.1  
B80.8  
F6.4  
B38.2  
F15.5  
B40.3  
F2.4  
B46.3  
F2.2  
B24.8  
F1.8  
B31.5  
F1.7  
B30.1  
F1.7
```

Feed/bend parameters

```
G1 Y7.6 F500  
G1 Y-3.6 F500  
G1 X2 F200  
G1 Y-1 F200  
G1 X1 F200  
G1 Y-1 F200  
G1 X1 F200  
G1 Y-1 F200  
G1 X2 F200  
G1 Y-1 F200
```

G-Code

Methods

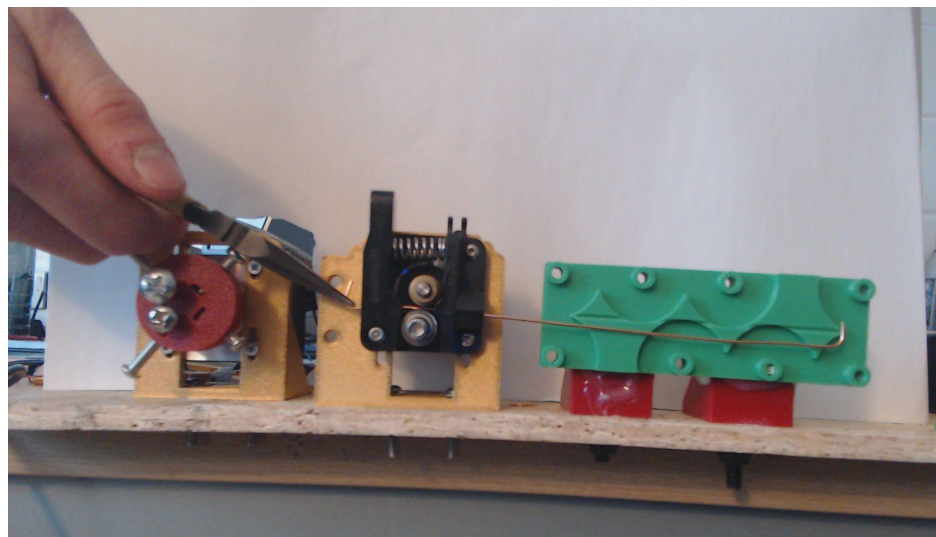


photograph

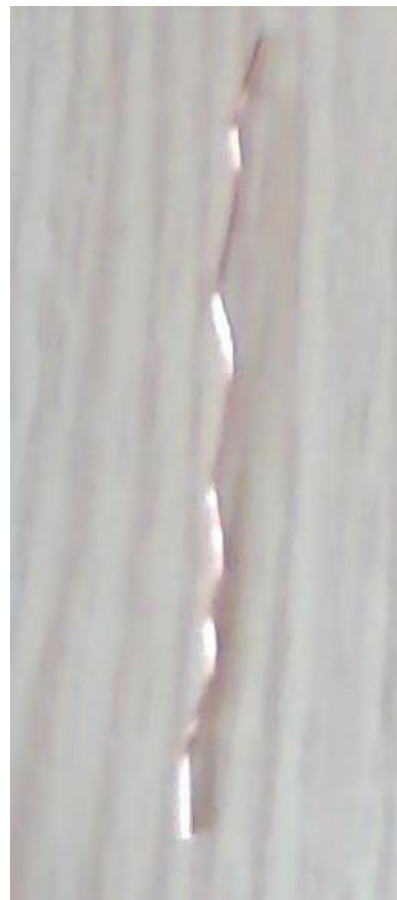
3D model

contour generation

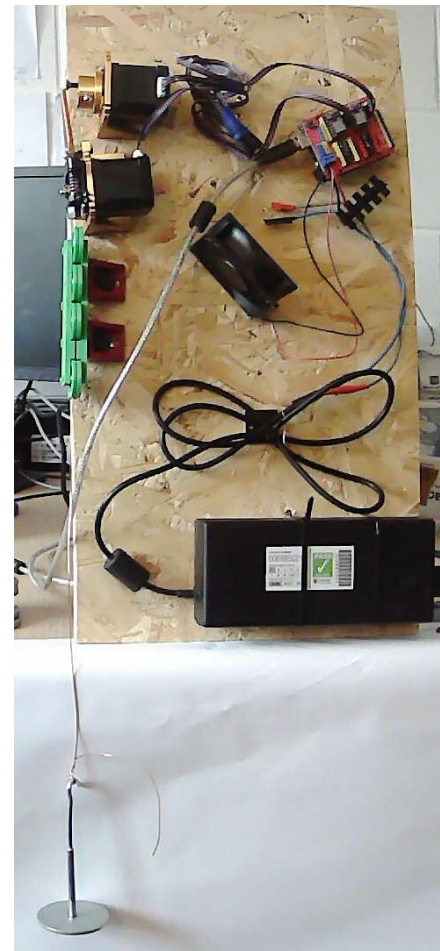
G-code



preknurling

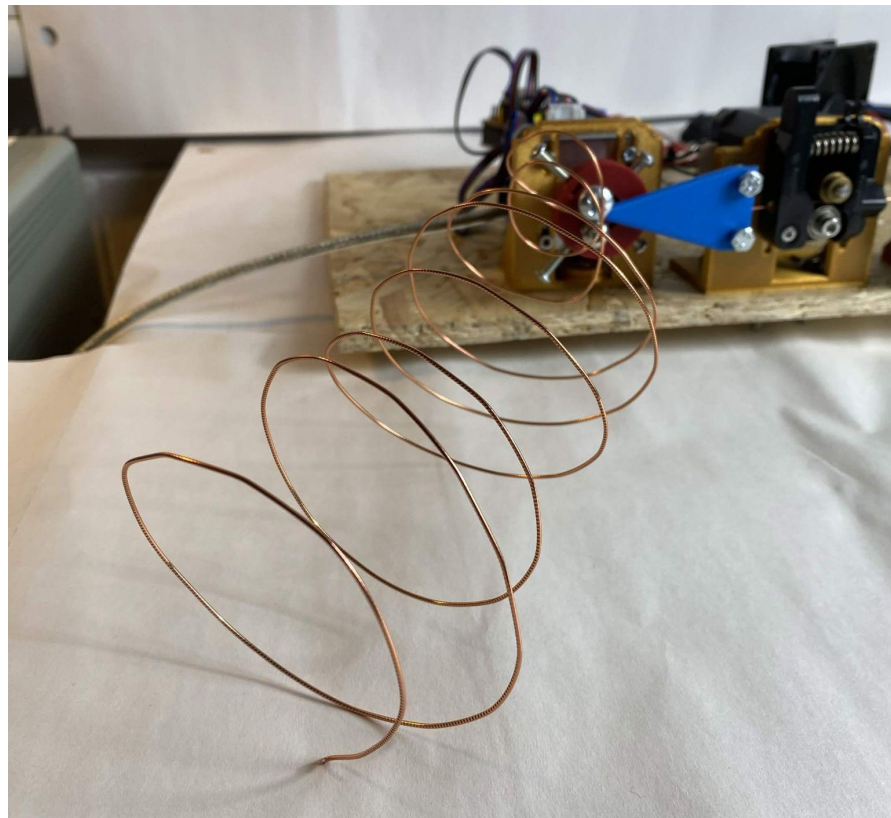
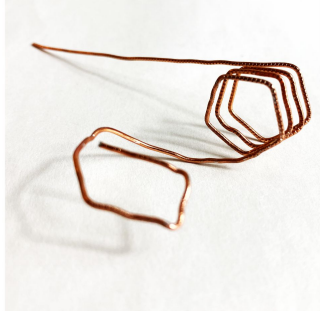
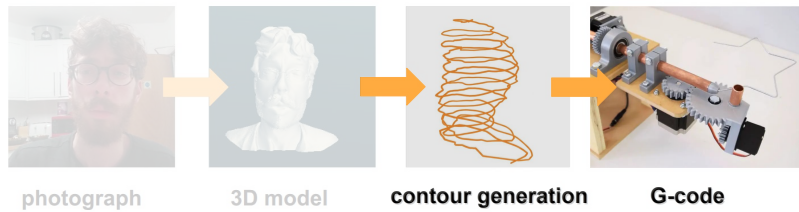


prekinking



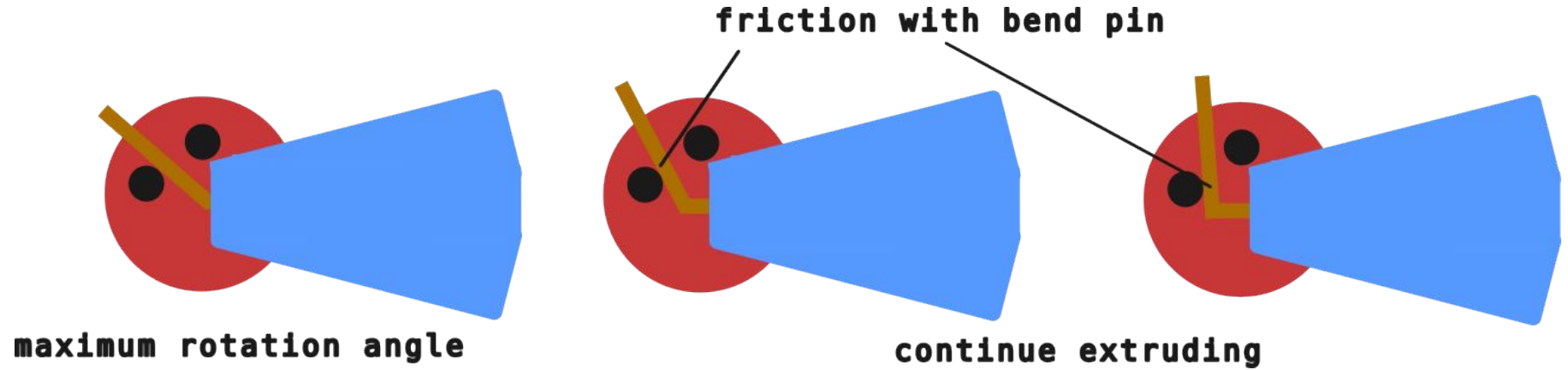
weight testing

Methods



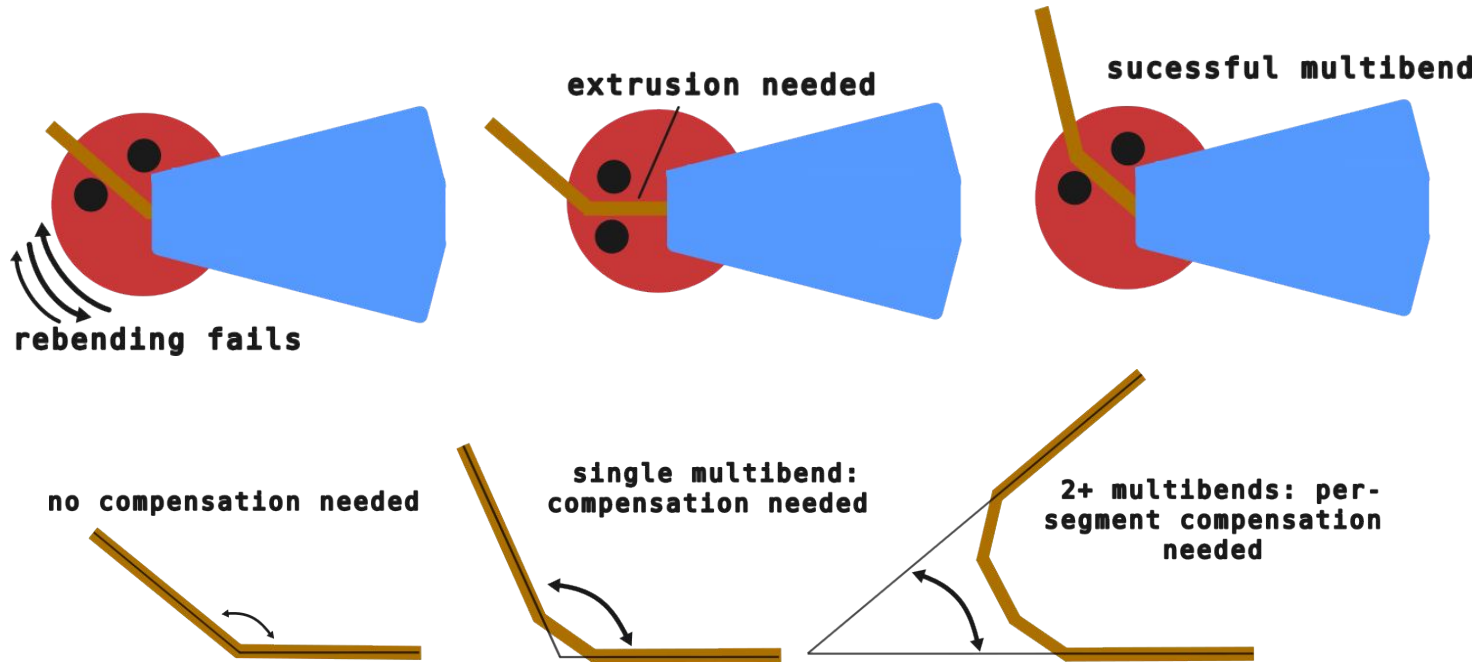
Contributions

Characterizing stick-slip friction as mechanism for wirebending



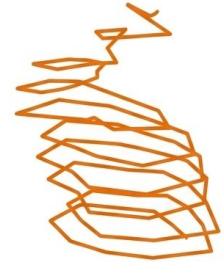
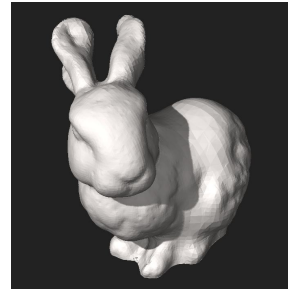
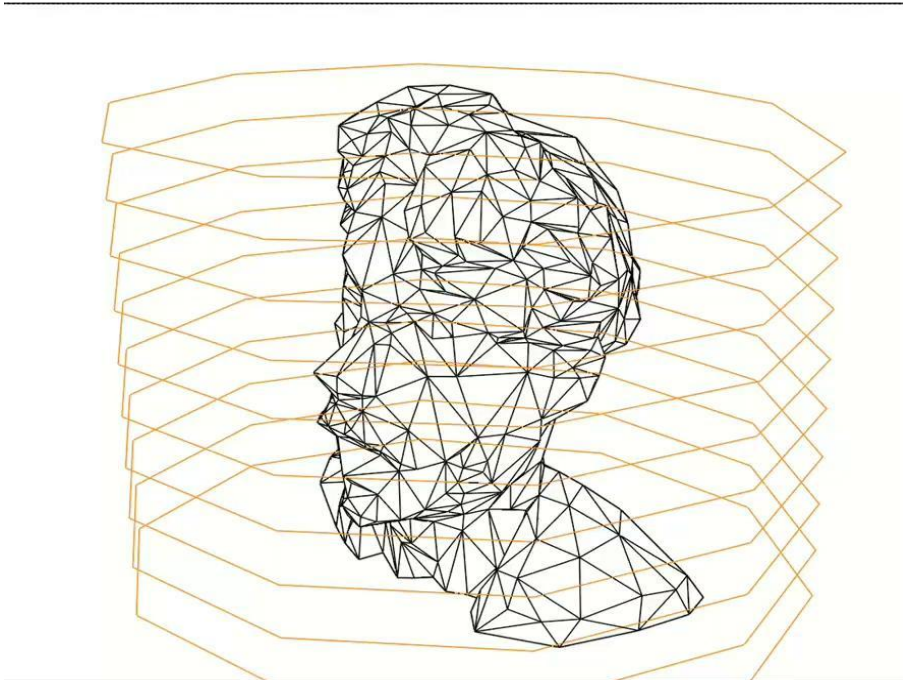
Contributions

Creation of a multibend compensation algorithm



Contributions

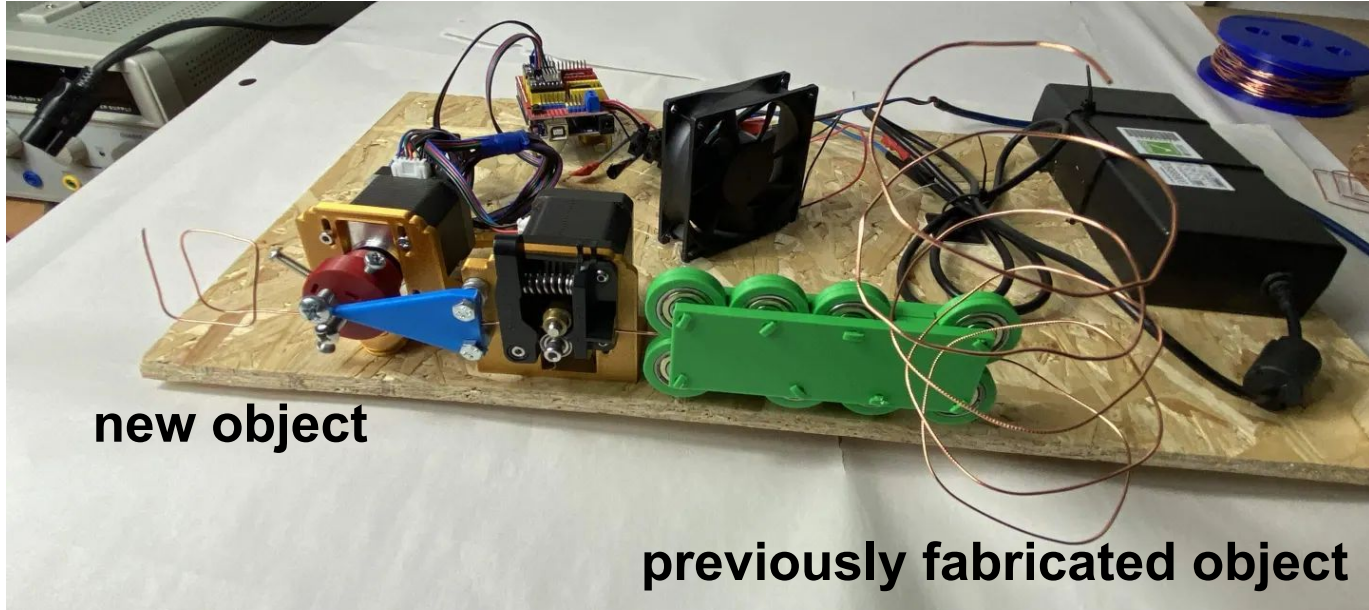
Creation of the surface-adaptive spiral wrapping algorithm



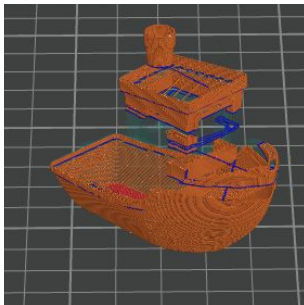
poor generation of non-continuous
cross-sections

Contributions

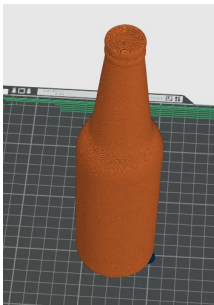
Demonstration of ephemeral fabrication



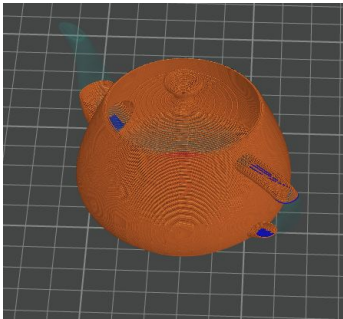
Conclusions



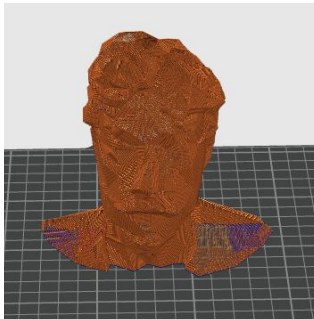
Benchy
20m



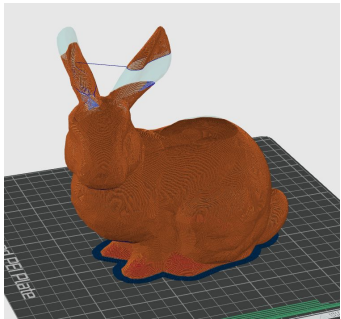
Beer Bottle
1h 39m



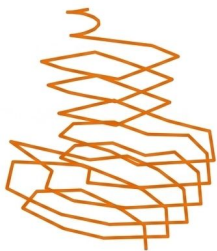
Utah Teapot
30m



Statue
1h 9m



Stanford Bunny
1h 18m



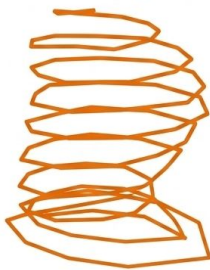
Benchy
7m



Beer Bottle
8m



Utah Teapot
7m



Statue
10m



Stanford Bunny
7m

Wirebending is found to be 5.8x faster and 4.7x more energy efficient than 3D printing